DO NOW — RECALL

Lesson 080 · Insertion Sort: Grow the Sorted Prefix

Name: _____ Class: _____ Date: _____

1. What gets fixed after a bubble pass?

2. What does insertion-sort key mean?

3. What does sorted prefix mean?

4. BST root10 with right child15: insert key12. Show comparisons and position.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Largest remaining value at end.

2. Model answer

Item currently being placed.

3. Model answer

Initial already-ordered portion.

4. Model answer

12>10 so go right;12<15 so insert as left child of15.

By the end, you can…

• The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

• Shift larger elements one position to the right, then insert key in the vacant position.

• Worst $O(n^2)$, already-sorted best O(n), auxiliary O(1); especially useful on small/nearly sorted arrays.

Vocabulary — know these terms

insertion sort — Insert key into ordered prefix.

key — Value currently being inserted.

shift — Move existing value to make room.

sorted prefix — Already-sorted left section.

NEW LEARNING

Read, annotate and explain

Understand

The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

Trace

Shift larger elements one position to the right, then insert key in the vacant position.

Explain

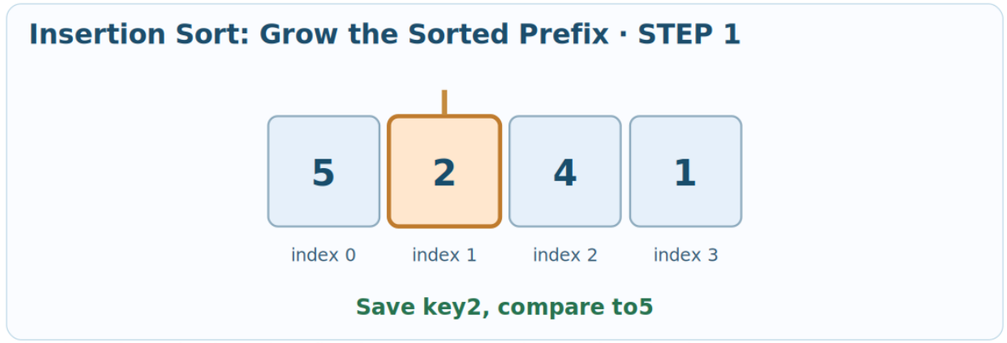
Worst $O(n^2)$, already-sorted best $O(n)$, auxiliary $O(1)$; especially useful on small/nearly sorted arrays.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

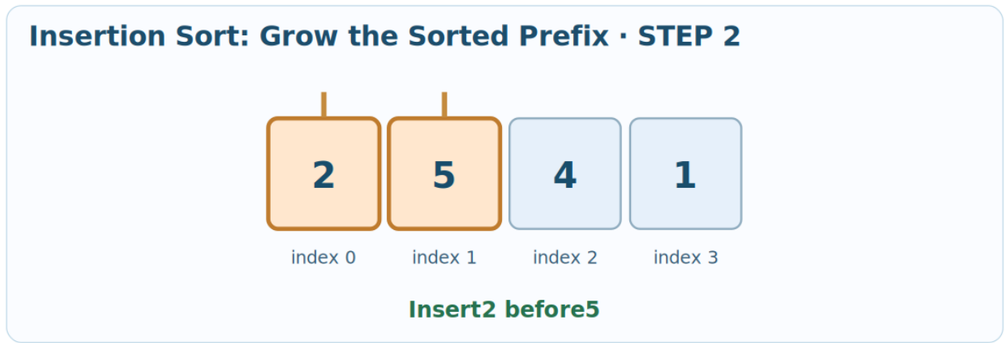
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Insertion Sort: Grow the Sorted Prefix · STEP 1; array 5, 2, 4, 1; Save key2, compare to5

New Learning



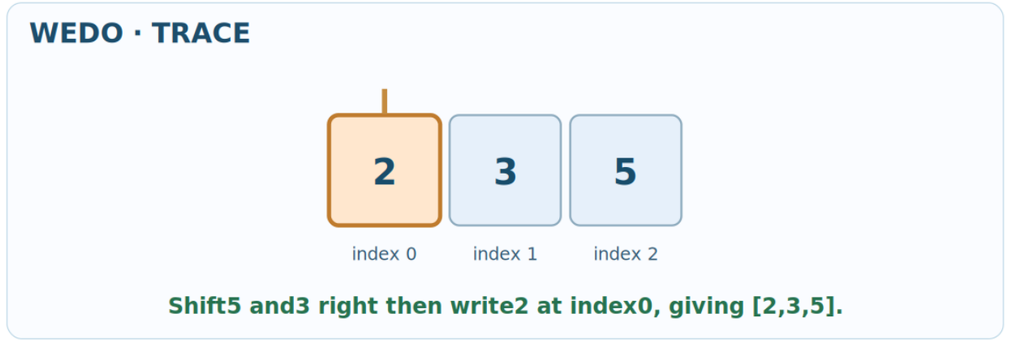
Insertion Sort: Grow the Sorted Prefix · STEP 2; array 2, 5, 4, 1; Insert2 before5

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert key2 in sorted prefix [3,5] of [3,5,2].

Shift5 and3 right then write2 at index0, giving [2,3,5].



WEDO · TRACE; array 2, 3, 5; Shift5 and3 right then write2 at index0, giving [2,3,5].

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

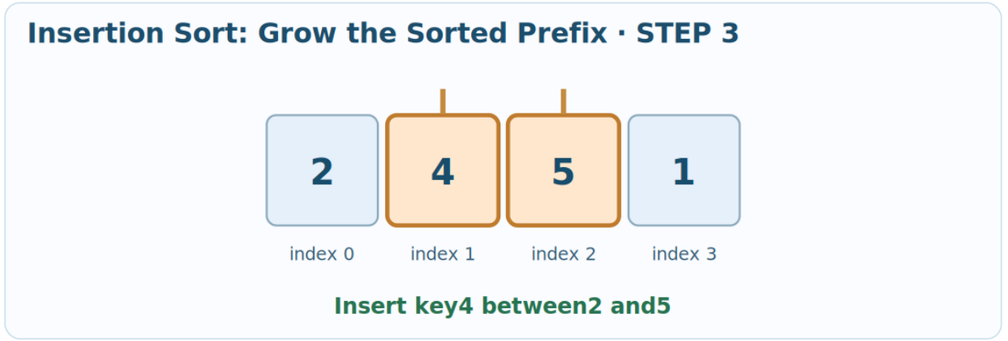
Guided 1 — CHECK

Check the working
Shift5 and3 right then write2 at index0, giving [2,3,5].

New Learning · original lesson diagrams

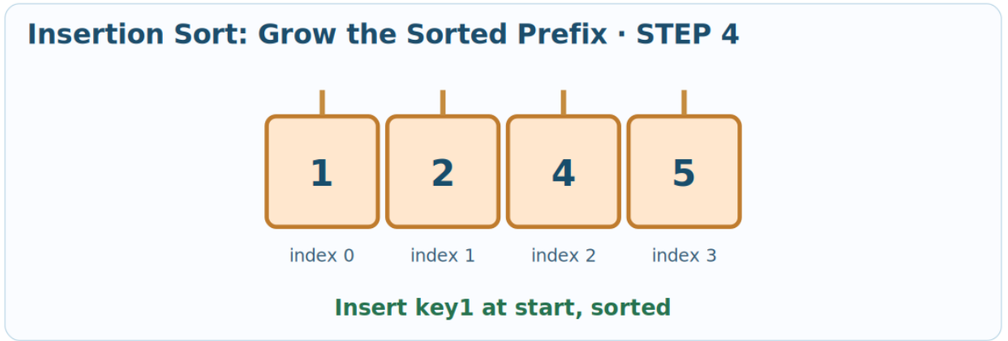
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Insertion Sort: Grow the Sorted Prefix · STEP 3; array 2, 4, 5, 1; Insert key4 between2 and5

New Learning



Insertion Sort: Grow the Sorted Prefix · STEP 4; array 1, 2, 4, 5; Insert key1 at start, sorted

I DO — WORKED EXAMPLE

Follow the worked method

Insertion-sort [4,1,3]; give states after each key.

Check the working

Key1→[1,4,3]; key3→[1,3,4].

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Insertion sort always requires a separate second array to store sorted values.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

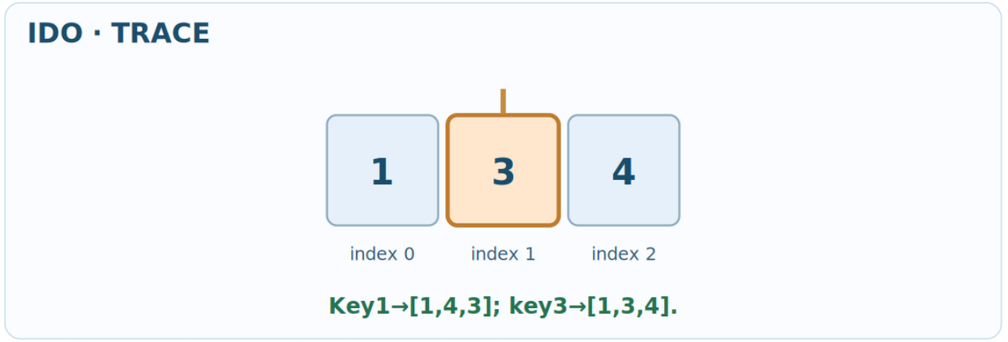
Compare insertion and bubble sort method/time on already sorted input. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insertion-sort [4,1,3]; give states after each key.

Key1→[1,4,3]; key3→[1,3,4].



IDO · TRACE; array 1, 3, 4; Key1→[1,4,3]; key3→[1,3,4].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Insert key2 in sorted prefix [3,5] of [3,5,2].
Source code / input:
Shift5 and3 right then write2 at index0, giving [2,3,5].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Give arrays after first two key insertions into [5,2,4,1]. [6]

Question 2

Complete insertion of final key1 into [2,4,5,1], showing working. [6]